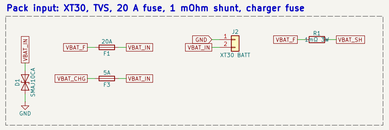
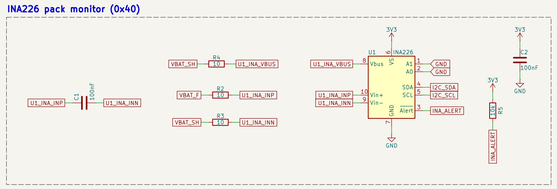
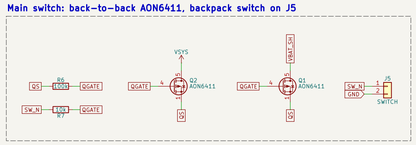
Pack input: XT30, TVS, 20 A fuse, 1 mOhm shunt, charger fuse



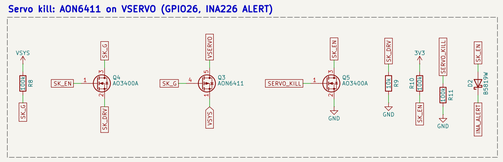
INA226 pack monitor (0x40)



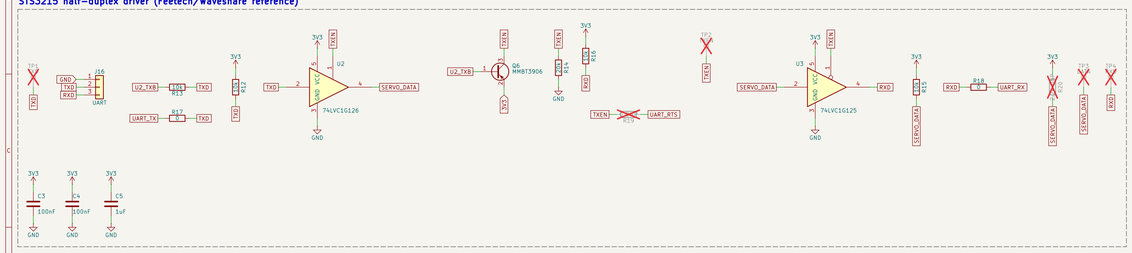
Main switch: back-to-back AON6411, backpack switch on J5



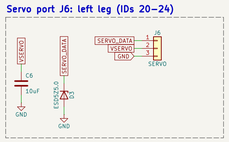
Servo kill: AON6411 on VSERVO (GPIO26, INA226 ALERT)



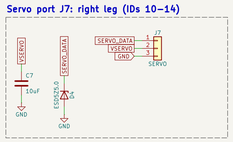
STS3215 half-duplex driver (Feetech/Waveshare reference)



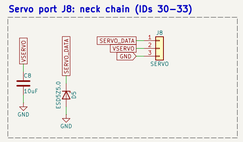
Servo port J6: left leg (IDs 20–24)



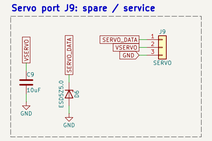
Servo port J7: right leg (IDs 10–14)



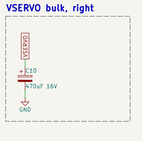
Servo port J8: neck chain (IDs 30–33)



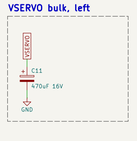
Servo port J9: spare / service



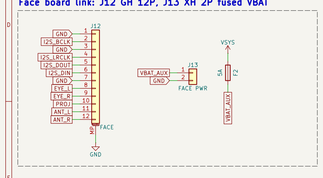
VSERVO bulk, right



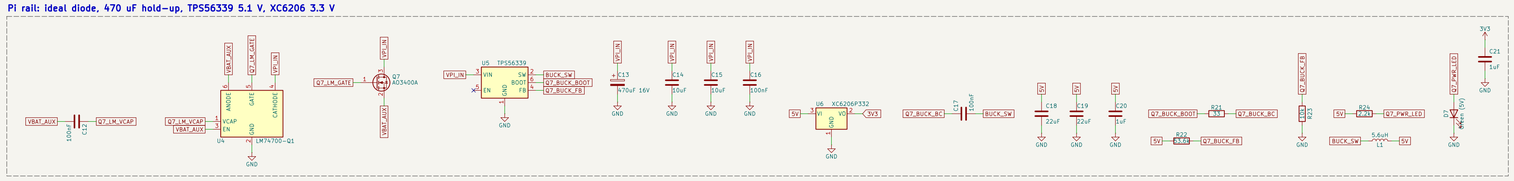
VSERVO bulk, left



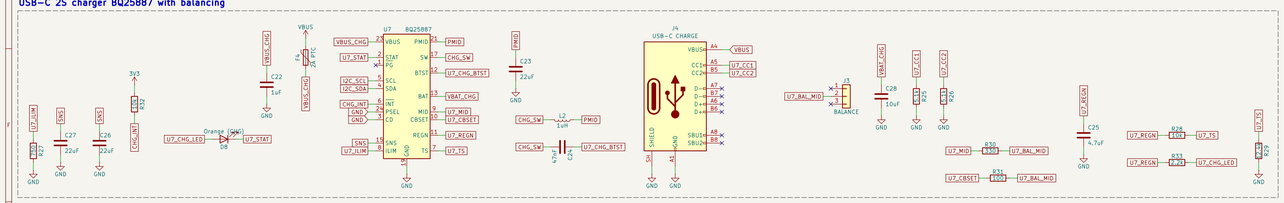
Face board link: J12 GH 12P, J13 XH 2P fused VBAT



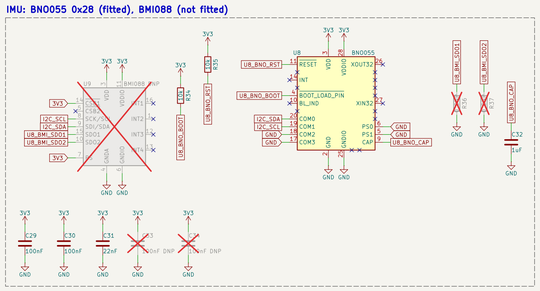
PI rail: ideal diode, 470 uF hold-up, TPS56339 5.1 V, XC6206 3.3 V



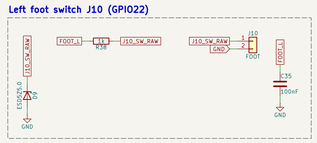
USB-C 2S charger BQ25887 with balancing



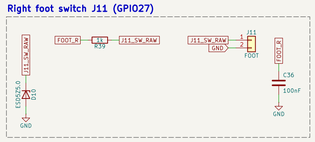
IMU: BNO055 0x28 (fitted), BMI088 (not fitted)



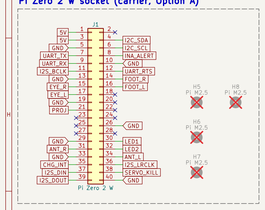
Left foot switch J10 (GPIO22)



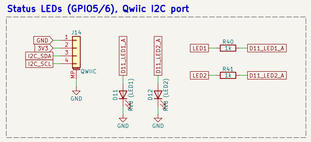
Right foot switch J11 (GPIO27)



Pi Zero 2 W socket (carrier, Option A)



Status LEDs (GPIO5/6), Qwiic I2C port



M2.5 holes on trunk_top's adapter bosses



Power flags



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Sheet: /
File: quack_core.kicad_sch

Title: Quack Core — brain board for the Open Duck Mini: Pi Zero 2 W carrier

Size: A1 | Date: 2026-10-03 | Rev: A
KiCad E.D.A. 10.0.6 | Id: 1/1